

The Swap-Refresh Gadgetization

local_mixing — design note

2026-08-20

Redesign of the single-carrier production gadgetizer (the `gen_sandwich_gadget` sandwich path): the aggregate-Gray fold is replaced by a per-gate *mask swap-with-refresh*, a *closing zero-slice block* is appended, and every measured output-port channel is closed. The result eliminates all detected linear (and degree-2) relations between the two circuits' wire segments *and* is about 30% smaller than the construction it replaces.

1 Goal

No GF(2)-linear equation should hold in which some variables are wire segments of the source (sandwich) circuit and others are wire segments of the gadgetized circuit — except through the public boundary (functions of the shared input at the input port, of the shared output at the output port), which any correct implementation exhibits.

The prior construction failed this in three measured ways.

1. **Endpoint cancellation.** Every logical value decodes as $v = c_v \oplus M_1 \oplus M_2 \oplus M_3 \oplus H \oplus \kappa$, the mask monomials M_i (degree 2) and H (degree 3) being products of *frozen* band literals. A fold writes the control's decode onto the target's carrier and never touches the target's own mask set, so the masks cancel in every before/after XOR across the gate:

$$\text{carrier}^+ \oplus \text{carrier}^- = \text{src}^+ \oplus \text{src}^-,$$

an exact cross-circuit linear identity — measured by `flip_match` at 100% of linear source gates and 32.4% of nonlinear ones.

2. **The Gray operand witness.** The aggregate-Gray fold gathers an operand's complete mask sum onto an accumulator wire and strips it back, so the operand is linearly reconstructible from three gadget segments: $B = c_b \oplus u^- \oplus u^+ \oplus \kappa$.
3. **Port asymmetry.** The zero-slice guard existed only at the input port; a reverse evaluator met a structurally different entry.

2 Execution

2.1 Per-gate mask swap-with-refresh

At every fold, the *target* value and *one randomly chosen control* each retire one base-degree mask monomial and gain a freshly drawn one. The target-side inject is emitted *strictly interior* to the

fold’s fragment stream; the control-side pair follows the fold, pinned by read/write collisions. No contiguous window of any carrier’s writes then XORs to a clean operand decode.

Two lemmas fixed the mechanism’s shape:

- **Verbatim moves are unsound.** Moving a monomial M between two mask sets emits it once on each touched wire, so over $\text{GF}(2)$ the two carriers’ deltas XOR back to the source delta: $\Delta(\text{t-carrier}) \oplus \Delta(\text{c-carrier}) = \Delta T$. Conservation makes any exact move leak; arriving monomials must be fresh draws.
- **Fresh means fresh band positions.** A polarity re-roll of the same product differs from it by a polynomial of degree $\leq \text{deg} - 1$ in the band wires, whose values are themselves wire segments — back inside a linear adversary’s span.

Swap mode **declines every Gray mode**: the gathers materialize an operand’s whole mask sum as an accumulator segment pair regardless of mask ownership, and the gather/strip snapshot cannot tolerate a mid-block registry change. The expanded fold is used instead, under a *target-stable commuting shuffle* that preserves per-wire write order (same-target XOR writes commute; the standard shuffle re-exposed a clean window at $\sim 14\%$ of arity-1 folds).

2.2 The closing zero-slice block

An independently drawn slice guard with the opening block’s specification — identity exactly on the zero band slice, every nonzero band slice perturbs the data — is appended after the mirror fill, so a reverse evaluator meets the same structure at their entry as a forward evaluator does. Its targets are confined to the low (forward-junk) data half: the honest forward run reaches the output port with a junked band, so the block fires there and must only perturb junk. The composite preserves the sandwich on the *upper data half only*, which is the payload contract.

2.3 Output-port hygiene

Reaching zero required closing four further channels, each found by re-running `flip_match` after the previous fix: the junk half is never stripped (bare junk segments hand their pair-XORs to the source’s own segment XORs); route-home runs *before* the strip, carrying the band-placement map, so routing swaps move masked carriers; the strip’s constant-discharge helper, both band fills, and the epoch refills source exclusively from never-bared wires (a refill copying a payload carrier into a band wire stores the payload bit where a half-captured emission window can read it back); and every relocation also refreshes one monomial of each moved value — a payload value is a fold target exactly once, so its mask function would otherwise recur identically at every relocation stop, and a segment pair cutting through two matching stops recovers the single value transition exactly.

3 Why it ended up smaller

The size reduction was not a goal; it fell out of declining Gray. The Gray fold was never cheap — it was bought for phase-A store-reachability (narrow fragments; $31.55\% \rightarrow 95.47\%$) at a measured $2.38\times$ the gates of the wide fold. Its per-gate machinery — gathering and stripping every mask atom of both operands onto two accumulators, plus the four-phase product schedule — costs more than the expanded fold’s fragment list at the production plan $[2, 2, 2, 3]$. The

swap-refresh construction removes all of that and spends back only four monomial emissions per fold, two per relocation, and one extra slice block:

n	prior (Gray)	swap-refresh	ratio
32	80,950	57,217	$0.71\times$
64	157,586	112,394	$0.71\times$
128	$\sim 1.29\text{M}$ (campaign builds)	878,863	$\sim 0.68\times$

The $n=128$ half-sandwich build ($|C|=|D|=3000$) comes to 481,070 gates.

The price is paid in a different currency: with the ladder off, $\sim 7\%$ of fold fragments remain wide, material phase A’s store cannot re-encode. Both measured remedies currently violate the zero-relation contract (`--prod-ladder-cap 4` re-opens a small linear tail through its borrowed scratch; `PROD_POST_FRAGMENT` a broad one) and are off-limits for deliverables until audited. Store reachability under the expanded fold is the open measurement for the next mixing campaign.

4 Verification ($n = 128$, production parameters, fresh random seed)

- **Function:** bit-sliced verify passed (payload half equals the sandwich on the honest slice); 317/317 unit tests.
- **Inputs $\leftrightarrow$ outputs:** the rank of the 257 columns $\{1, x, y\}$ over 1024 samples is full (257/257) — no affine relation between inputs and outputs (conclusive: a true relation forces a rank deficit on every sample set).
- **flip_match** (local segment-pair XOR vs per-gate source deltas, kwin 12/30/60): 0/13,568 source gates matched — down from 100% linear / 32.4% nonlinear on the prior build.
- **segment_deduce, degree 1** (8 cuts, 16,384 samples, fresh-sample verified): 2.91% of sandwich segments deducible, every equation on the shared-input or shared-output cut; *zero* interior-cut equations (prior build: $\sim 50\%$ deducible with thousands of interior equations in the gadget-exposure direction).
- **segment_deduce, degree 2** (four restricted windows up to 12,881 monomials, both directions, matched degree-1 controls): degree 2 adds exactly one equation — an early C-gate output that is a quadratic of the *public input*, matched on the input cut only. No other verified equation uses a product; the reverse direction’s interior-labeled hits are the public payload $C(x)$ at the sandwich’s C/D seam matching the gadget’s output, present for any correct implementation. Caveats: windows are restricted, and products are one-sided (no bilinear $s \cdot g$ cross-terms in the model).

5 Operation

`ProdConfig::production_single()` now sets `swap_refresh: 1, close_slice: 1`; environment overrides `PROD_SWAP / PROD_CLOSE_SLICE` (0/0 restores the prior Gray stream byte-for-byte). The `gss_mix` pipeline picks the new defaults up unchanged. The two-share (`sss`) and multi-carrier paths receive the fold-level swap wherever `fold_cg` runs but not the output-port hygiene; extending them is future work.